Data Dictionary

Table Name	Column Name	Data Type
users	id	BIGINT PK
	name	VARCHAR
	email	VARCHAR UNIQUE
	role	ENUM(admin, teacher, student)
	student_id	VARCHAR UNIQUE NULL
	academic_year	VARCHAR NULL
	created_at	TIMESTAMP
	updated_at	TIMESTAMP
categories	id	BIGINT PK
	name	VARCHAR UNIQUE
	description	TEXT NULL
	is_published	BOOLEAN
	time_limit_minutes	UNSIGNED INT
quizzes	id	BIGINT PK
	title	VARCHAR
	category_id	BIGINT FK
	teacher_id	BIGINT FK NULL
	difficulty	ENUM(Easy, Medium, Hard)
	duration_minutes	UNSIGNED INT
	is_active	BOOLEAN
questions	id	BIGINT PK
	quiz_id	BIGINT FK NULL
	category_id	BIGINT FK
	question_text	TEXT/LONGTEXT
	question_type	ENUM(mcq, tf, multi_select, ordering, short_answer)
	points	INT
question_options	id	BIGINT PK
	question_id	BIGINT FK
	option_text	VARCHAR
	is_correct	BOOLEAN
	order_index	INT NULL
answers	id	BIGINT PK
	question_id	BIGINT FK
	answer_text	VARCHAR
	is_correct	BOOLEAN
quiz_attempts	id	BIGINT PK
	student_id	BIGINT FK
	quiz_id	BIGINT FK
	status	VARCHAR
	score_percent	DECIMAL(5,2)
	attempt_type	VARCHAR

Normalization (1NF, 2NF, 3NF)

1NF: Core columns are atomic; JSON is limited to flexible payloads.

2NF: Composite business facts are in bridge tables (attempt_answers, retake allowances).

3NF: Main entities are separated; quiz_attempt aggregates are deliberate performance fields.

Notes

- questions has both category_id and quiz_id; keep them synchronized in logic.
- answers and question_options overlap in correctness modeling; keep one clear write path.